AP BIOLOGY TOPIC 6.8

Biotechnology: Gel Electrophoresis, PCR, Plasmids & Sequencing

UNIT 6: GENE EXPRESSION & REGULATION (12–16% WEIGHT)

IST-1.P | GEL ELECTROPHORESIS | PCR (TAQ POL) | RECOMBINANT PLASMIDS

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 6.8 (BIOTECHNOLOGY: GEL ELECTROPHORESIS, PCR, PLASMIDS & SEQUENCING)

CED Core Vocabulary

Restriction Enzymes (Endonucleases) — Bacterial enzymes (e.g., EcoRI, BamHI) that recognize specific short palindromic DNA sequences (4–8 bp) and cleave phosphodiester bonds to produce staggered...

Gel Electrophoresis Principles — Separates DNA, RNA, or protein fragments through a porous agarose/polyacrylamide gel matrix based on charge and molecular size (length in base pairs, bp) .

Size Sieving Rule — Smaller (shorter) DNA fragments navigate the gel pores easily and migrate FASTER and FARTHER from the wells; larger fragments move slowly and stay near the top.

Purpose — Rapidly amplifies a specific target DNA region into millions/billions of copies (2^n after n cycles) in a thermal cycler without needing living cells.

4 Required Reaction Components — (1) Target template DNA, (2) Forward & Reverse DNA primers (flank target sequence), (3) Free deoxynucleotides (dNTPs: A, T, C, G), and (4) Heat-stable Taq...

Three Repeating Thermal Steps per Cycle — 1. Denaturation (~94–96°C): High heat breaks hydrogen bonds to separate double-stranded DNA into single strands (replaces Helicase!).

2. Annealing (~50–65°C) — Reaction cools so forward and reverse DNA primers hydrogen-bond to complementary 3′ ends of the target sequence.

3. Extension / Elongation (~72°C) — Optimal temp for Taq polymerase to synthesize new complementary DNA strands 5′ → 3′ starting from the primers.

§2. RESTRICTION ENZYMES & GEL ELECTROPHORESIS

Size & Charge Separation

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Restriction Enzymes & Gel Electrophoresis ➔

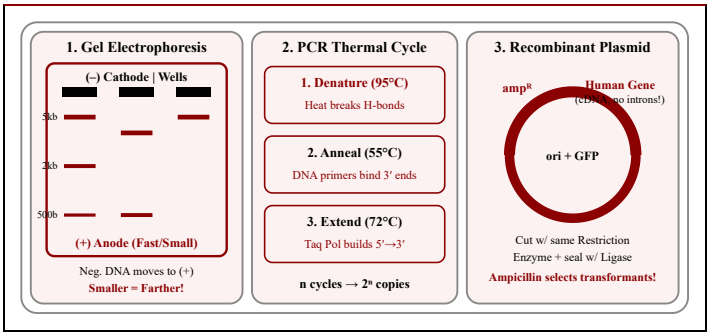
Restriction Enzymes (Endonucleases) ➔ **Gel Electrophoresis Principles**

- **Restriction Enzymes (Endonucleases):** Bacterial enzymes (e.g., *EcoRI, BamHI*) that recognize specific short **palindromic DNA sequences** (4–8 bp) and cleave phosphodiester bonds to produce staggered single-stranded overhangs called **"sticky ends"** (or blunt ends).
- **Gel Electrophoresis Principles:**
  - Separates DNA, RNA, or protein fragments through a porous agarose/polyacrylamide gel matrix based on **charge and molecular size (length in base pairs, bp)**.
  - **Why DNA Moves Toward the Positive Electrode (Anode, +):** DNA has a uniform **net negative charge** due to the phosphate groups (**$-PO_4^{2-}$**) in its sugar-phosphate backbone. DNA is loaded into wells at the **negative cathode (–)** and migrates toward the **positive anode (+)** ("*Run to Red*").
  - **Size Sieving Rule: Smaller (shorter) DNA fragments** navigate the gel pores easily and **migrate FASTER and FARTHER** from the wells; **larger fragments** move slowly and stay near the top.
- **Applications (DNA Fingerprinting / RFLP / STR Profiling):** Forensics crime-scene matching, paternity testing (child's bands must be a 50/50 combination of mother + biological father), and evolutionary/medical allele diagnostics.

[CORE AXIOM / REMEMBER] **Gel Electrophoresis Rule:** DNA is **negatively charged (** **PO_4^{2-}** **)** → moves toward the **positive (+) pole**; **Smaller fragments travel FARTHER!** (Distance traveled is inversely proportional to **$\log_{10}$(bp)**).

§4. BIOTECHNOLOGY LAB TOOLKIT VISUALIZED

Gel • PCR • Transformation



§5. RECOMBINANT DNA, BACTERIAL TRANSFORMATION & SEQUENCING

Cloning & CRISPR

- **Building Recombinant DNA & Bacterial Transformation:**
  - **1. Cut with the SAME Restriction Enzyme:** Cut both the bacterial **plasmid vector** and the foreign DNA (gene of interest) with the *exact same restriction enzyme* so their complementary sticky ends hydrogen-bond together.
  - **2. Ligate with DNA Ligase:** Add **DNA Ligase** to form covalent phosphodiester bonds, sealing the foreign gene into the recombinant plasmid.
  - **3. Why Use Eukaryotic cDNA (Complementary DNA)?** Because bacteria lack spliceosomes and **cannot remove introns**, scientists use *reverse transcriptase* on mature eukaryotic mRNA to make intron-free **cDNA** (plus a **bacterial promoter**) before inserting a human gene (like **human insulin**) into *E. coli*!
  - **4. Selection on Antibiotic Plates (** **amp^R** **/ GFP /** **lacZ** **):** Heat-shock **$CaCl_2$** –treated competent bacteria and plate on agar containing **ampicillin**. Only bacteria that successfully took up the plasmid carrying the **ampicillin-resistance gene (** **amp^R** **)** survive and form colonies!
- **DNA Sequencing & CRISPR-Cas9 Genome Editing:**
  - **DNA Sequencing:** Determines the exact nucleotide order (**A, T, C, G**) of a DNA fragment to identify disease mutations, build phylogenetic trees, and map genomes.
  - **CRISPR-Cas9:** Uses a synthetic **guide RNA (gRNA)** complementary to a target DNA sequence to direct the **Cas9 endonuclease** to cut a specific genomic locus to knock out or repair a gene.

[MECHANISTIC NOTE] **Circular vs. Linear Restriction Digest Math on AP Exams:**

- Cutting a **LINEAR** DNA molecule **n** times produces **n + 1** **fragments (bands)**.
- Cutting a **CIRCULAR plasmid** **n** times produces **n** **fragments (bands)**!

§3. PCR (POLYMERASE CHAIN REACTION): IN VITRO DNA AMPLIFICATION

2^n Exponential Copies

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

PCR (Polymerase Chain Reaction) ➔ **Purpose** ➔
4 Required Reaction Components ➔ **Three Repeating Thermal Steps per Cycle**

- **Purpose:** Rapidly amplifies a specific target DNA region into **millions/billions of copies (2^n after n cycles)** in a thermal cycler without needing living cells.
- **4 Required Reaction Components:** (1) Target template DNA, (2) Forward & Reverse **DNA primers** (flank target sequence), (3) Free deoxynucleotides (**dNTPs: A, T, C, G**), and (4) Heat-stable ***Taq* DNA Polymerase** (isolated from thermophilic bacterium *Thermus aquaticus* so it doesn't denature at 95°C!).
- **Three Repeating Thermal Steps per Cycle:**
  1. **1. Denaturation (~94–96°C):** High heat breaks **hydrogen bonds** to separate double-stranded DNA into single strands (replaces Helicase!).
  2. **2. Annealing (~50–65°C):** Reaction cools so forward and reverse **DNA primers hydrogen-bond** to complementary 3′ ends of the target sequence.
  3. **3. Extension / Elongation (~72°C):** Optimal temp for ***Taq* polymerase** to synthesize new complementary DNA strands **5′ → 3′** starting from the primers.

[AP EXAM FRQ SCORING STRATEGY] Why PCR Uses Heat Instead of Helicase & DNA Primers Instead of Primase: Heating to 95°C denatures H-bonds physically (and would denature human DNA polymerase, which is why thermostable ***Taq* polymerase** is required), while synthetic DNA primers are added directly to the tube!

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 6.8 [MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]

Topic 6.8: Biotechnology: Gel Electrophoresis, PCR, Plasmids & Sequencing ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔
Experimental Perturbation (±2 SE_x̄) ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 8 & 9 (pGLO Bacterial Transformation & Restriction Enzyme Analysis of DNA): Heat-shock transformation of E. coli with pGLO plasmid (bla ampicillin-resistance gene + GFP under arabinose-inducible araC promoter). Plate comparisons: −pGLO/LB (lawn, viability control), −pGLO/LB+Amp (no growth, antibiotic lethality control), +pGLO/LB+Amp (white colonies, transformation), +pGLO/LB+Amp+Ara (glowing green colonies under UV, inducible gene expression). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative & Graph Analysis: Transformation Efficiency = (Total colonies counted on selective plate) / (µg of plasmid DNA spread). Gel Electrophoresis Semi-Log Standard Curve: DNA fragment size (bp, $\log_{10}$ scale on y-axis) vs. migration distance (mm, linear scale on x-axis) yields a straight line for interpolating unknown fragment lengths. PCR amplification after n thermal cycles yields 2^n target DNA copies. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Molecular Disruption Case Study: Splice-site mutations in the β-globin gene (β-Thalassemia) prevent the spliceosome (snRNPs) from excising an intron or cause exon skipping, shifting the mRNA reading frame and introducing a premature nonsense stop codon (UAA/UAG/UGA). In oncology, epigenetic hypermethylation of CpG islands in tumor-suppressor promoters (e.g., BRCA1 or MLH1) or histone deacetylation condenses euchromatin into transcriptionally silent heterochromatin without altering the primary DNA sequence.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 6.8 to prove why the claim holds.

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